

KEVIN KIRUI

Healthcare AI & Workflow Systems Specialist

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Portfolio: <https://kevin-kirui.vercel.app/>

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GitHub: <https://github.com/arapkirui513-hub>

PROFILE

Healthcare AI and workflow systems specialist with a biomedical engineering background and 2+ years spanning clinical engineering, healthcare data, AI evaluation, and software projects.

Builds healthcare applications using Python, FastAPI, React, and SQL; evaluates AI-generated responses; validates healthcare data; and translates clinical operations problems into structured software and analytics solutions.

Built 4 public healthcare applications, including a dashboard analyzing **10,483 health facilities across 47 Kenyan counties**.

WORK

Kenya Health Facilities Dashboard

Healthcare Analytics & Public Health Data

- Built and deployed a healthcare analytics application using Python, FastAPI, React, and health facility data covering **10,483 facilities across all 47 Kenyan counties**.
- Developed facility distribution, service availability, and healthcare gap analysis capabilities.
- Improved mobile PageSpeed from **56 to 95** while maintaining **100 scores for Accessibility, Best Practices, and SEO**.
- Reduced initial API requests from **6 to 1** through a combined dashboard-data endpoint.
- Added API documentation and CSV export functionality.

Clinical Workflow Signal Audit

Healthcare Workflow Intelligence

- Designed a synthetic ICU workflow audit using clinical signals and signal-to-action latency tracking to identify workflow risks and escalation delays.
- Built **Monitor, Review, Escalate, and Activate** tiers with role-based review.
- Structured workflow events into audit trails and SLA-based escalation routing.

Remote HealthAI Role Matcher

Research & Decision-Support System

- Built a repeatable workflow for identifying and prioritizing remote healthcare AI opportunities using scoring, filtering, and ranking logic.
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PROFESSIONAL EXPERIENCE

Healthcare AI & Workflow Systems

Self-Directed Development & AI Evaluation

May 2024 - Present

- Build and deploy healthcare analytics, AI evaluation, and workflow intelligence applications using Python, FastAPI, React, and modern web tools.
 - Evaluate AI-generated responses for accuracy, instruction-following, tone, usefulness, and healthcare safety across healthcare-related tasks.
 - Translate clinical and operational problems into structured datasets, workflow logic, dashboards, and decision-support interfaces.
 - Document technical work through GitHub, READMEs, and API documentation.
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HEALTHCARE EXPERIENCE

Kisii Teaching & Referral Hospital

Biomedical Engineering / Technical Support

Feb 2023 - Apr 2024

- Supported medical equipment maintenance, inspection, troubleshooting, and technical documentation across hospital departments.
- Worked alongside clinical teams to develop practical understanding of equipment availability, workflow dependencies, and downtime.

Kenyatta National Hospital & Tenwek Hospital

Biomedical Engineering Clinical Exposure

Gained clinical-environment exposure across ICU, renal, laboratory, diagnostic, and general hospital equipment workflows.

CORE SKILLS

Programming & Data

Python · SQL · Pandas · NumPy · Excel · Google Sheets · Data Cleaning · Data Validation · CSV Processing

AI & Workflow

LLM Response Evaluation · Model Output Review · Data Annotation · Instruction-Following · Healthcare Safety · Clinical Workflow Analysis

Software & Product

FastAPI · React · Tailwind CSS · GitHub · Vercel · API Documentation · Frontend/Backend Testing

Research & Documentation

Workflow Documentation · Structured Research · Technical Reporting · Markdown · Notion

EDUCATION

Kenya Medical Training College, Nairobi

Diploma in Biomedical Engineering

February 2026

Kenya Medical Training College

Certificate in Medical Engineering

December 2023
